Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	6
Topic	Construction
Subtopic	Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees
Theme	—
Curriculum objectives	• construct parallel lines • construct perpendicular lines • bisect a given line segment • construct angle 90° • construct angle 60°
Previous knowledge	• Students have been taught how to draw lines.
Lesson sequence	• construct parallel lines — the teacher draws the construction of parallel lines on the board step by step, using a set square and ruler and explaining aloud as she draws; the student construct parallel lines, following the teacher's board construction and replicating it in their notes • construct perpendicular lines — the teacher draws perpendicular lines on the board step by step, using a set square and ruler and explaining aloud as she draws; the student follow the teacher's board construction and replicate it in their notes • bisect a given line segment — the teacher draws the bisection of a line segment on the board step by step with a pair of compasses, explaining aloud as she draws; the student construct line segment, following the teacher's board construction and replicating it in their notes • construct angle 90° — the teacher constructs angle 90 degrees on the board step by step by compass arcs from a point on the line, explaining aloud as she constructs; the student construct angle 90 degrees, following the teacher's board construction and replicating it in their notes • construct angle 60° — the teacher constructs angle 60 degrees on the board step by step by compass arcs of equal radius from the endpoint, explaining aloud as she constructs; the student construct angle 60 degrees, following the teacher's board construction and replicating it in their notes
Instructional materials	• Mathematical set.
Evaluation	—

Lesson objectives

1.	construct parallel lines
2.	construct perpendicular lines
3.	bisect a given line segment
4.	construct angle 90°
5.	construct angle 60°

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees

Class: JSS1 **Subject:** Mathematics **Week:** 6

Section 1: Drawing a line parallel to a given line through a point

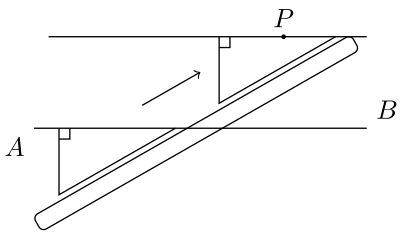
Two lines are **parallel** when they run side by side, the same distance apart all the way along, and never meet — like the two rails of a railway track. In construction we draw a line through a given point so that it stays the same distance from an existing line. The tool that does this for us is the **set square** working against a **ruler**: the ruler gives the slide a fixed direction, and the set square slides along it without turning, so the edge you draw through always keeps the same direction as the line you started with.

Here is the whole construction, on a line AB with a point P marked above it (fig-01).

1. **Lay the set square on AB.** Place the set square so that one of its edges lies exactly along the line AB.
2. **Put the ruler against another edge.** Place your ruler along one of the set square's other edges. If you are right-handed, use the left-hand edge. Now the set square cannot turn — it can only slide up and down the ruler (fig-02 and fig-03).
3. **Slide the set square up to P.** Hold the ruler firmly in place and slide the set square along it towards P. Stop when the edge that was on AB passes through P (fig-04).
4. **Draw the line.** Draw a line along that edge of the set square through P. Because sliding moved the set square without turning it, this new edge keeps exactly the same direction as it had on AB — so the line you draw through P is parallel to AB (fig-05).

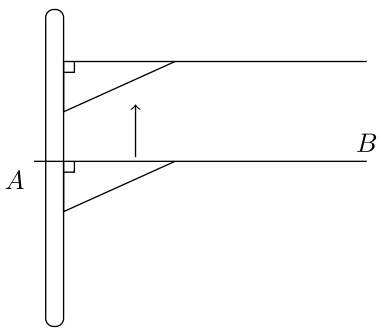
The set square is drawn at one size with its right angle at one end in the figures, but neither matters: any set square works, either way round, so long as one edge lies along AB and the ruler rests against another edge. The ruler's own angle does not affect the result either — it just steadies the slide.

Visual



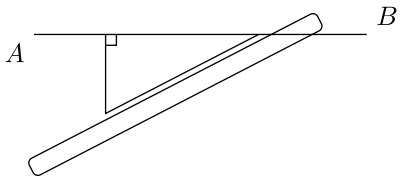
The same slide as fig-04, made against the leaning ruler of fig-03. The set square is drawn twice: once with its edge on AB, and once slid along the ruler until that edge passes through P. The two positions are congruent and point the same way — sliding along a held ruler moves the set square without turning it — so the edge through P stays parallel to AB. The arrow gives the direction of the slide, along the ruler's edge, and the right angle is marked in both positions.

Visual



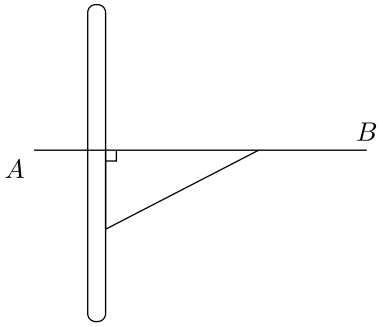
The vertical ruler held fixed, with the set square drawn twice — once with its edge on AB and once higher up, its edge on a second line parallel to AB — and an arrow between the two positions pointing up along the ruler.

Visual



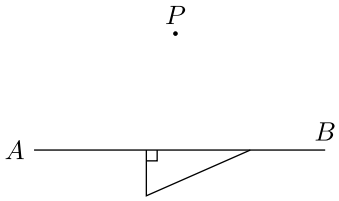
The same set square on AB as in fig-02, with the ruler moved to rest against its sloping edge instead of its upright one. The set square has not moved; only the ruler has, so it now leans across AB, running from lower left to upper right. The right angle is still where the edge on AB meets the short upright edge, and it is marked there. The ruler's inclination does not affect the construction. The set square is drawn at one particular size, and with its right angle at one particular end; neither is part of the method. Any set square works, either way round, so long as one edge lies along AB and the ruler rests against another.

Visual



The same set square on AB, with the ruler now drawn as a long strip lying along the set square's downward edge. The right angle marked at the corner is the set square's own, between the edge lying along AB and the edge running down from it.

Visual



Line AB, with a point P marked above it. A set square is drawn below AB with one edge along the line: the right angle is at the left end of that edge, one leg runs down from it, and the hypotenuse closes back to a point on AB to the right.

Worked example. Construct a line through point P parallel to the line CD.

Step 1. Place the set square on the line CD, one edge lying exactly along it.
Step 2. Hold the ruler against the set square's other edge (the left-hand edge if you are right-handed), so the set square is locked to the slide.
Step 3. Hold the ruler firmly and slide the set square along it towards P, until the edge that was on CD reaches P.
Step 4. Draw a line along that edge through P.

The line you have drawn through P keeps the direction of CD, because the set square slid without turning. It is parallel to CD.

Practice

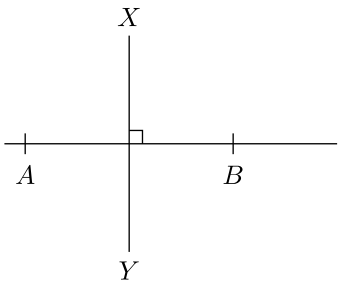
1. Draw a horizontal line PQ about 8 cm long and mark a point R about 3 cm above it. Construct a line through R parallel to PQ. Describe the steps you used.
2. In the construction of a line through a point parallel to a given line, why must you hold the ruler firmly while you slide the set square along it? What would happen if the ruler moved?

Section 2: Constructing Perpendicular Lines

Two lines are said to be **perpendicular** to each other when they meet at a right angle, that is, at 90°. Look at fig-06: line AB has a second line XY crossing it, with X above and Y below, and the little square at the crossing marks the right angle. Whenever two lines meet and the angle between them is a right angle, the lines are perpendicular.

A set square has one corner that is exactly a right angle. Because of this, you can use it, together with a ruler, to draw a line perpendicular to another line through any point on it.

Visual



Line AB with a second line XY crossing it, X above and Y below, and a right-angle mark in the upper right quadrant at the crossing.

Let's construct a line perpendicular to AB through a point Q on it.

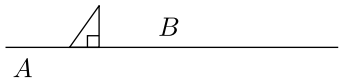
Step 1. Draw the line AB and mark a point Q on it, as in fig-07.

Step 2. Place the edge of the set square on line AB, so that its base lies along the line. Let the edge of the ruler meet the hypotenuse of the set square, as shown in fig-08 — the set square stands on AB like a narrow right-angled triangle, with the upright edge rising from the line and the right angle marked there.

Step 3. Pressing firmly on the ruler, slide the set square down along the hypotenuse side until the third edge touches point Q. Then draw a line XY along that third edge. Because the set square's upright edge is a right angle, the line you draw crosses AB at a right angle.

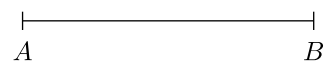
Fig-09 shows the result: line XY crosses AB with X above and Y below, and the right-angle mark in the upper right quadrant confirms the two lines meet at 90°. The line XY is perpendicular to AB.

Visual



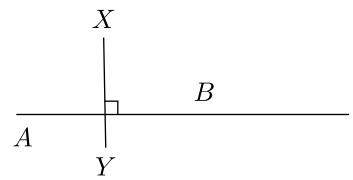
Line AB with the set square standing on it as a narrow right-angled triangle: a short base along AB, the upright edge rising from the right end of that base with the right angle marked there, and the hypotenuse closing back down to the left.

Visual



A plain line with a stroke at each end, labelled A on the left and B on the right.

Visual



Line AB with XY now drawn across it, X above and Y below, and a right-angle mark in the upper right quadrant.

Practice

1. Two lines are perpendicular when they meet at a right angle. What is the size, in degrees, of the angle between two perpendicular lines?
2. Using your set square and ruler, construct a line perpendicular to the line AB through a point Q on it. Name the perpendicular line XY.

Section 3: Bisecting a Line Segment

To bisect a line segment means to divide it into two equal halves. When you bisect a segment AB, you find the point M exactly at its middle, so that AM = MB. The construction uses a pair of compasses and a ruler, and the line that crosses AB meets it at a right angle (90°). The two crossing points you make with the compasses are labelled P and Q, and joining them gives you the line that cuts AB in half.

Follow these steps to bisect the line segment AB (fig-10 shows the starting segment, with A on the left and B on the right).

Step 1. Open the pair of compasses so that the radius is about ¾ of the length of AB.

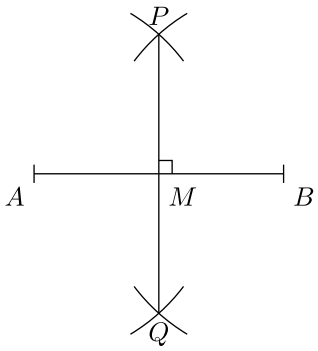
Step 2. Place the sharp point of the compasses on A and draw two arcs — one above and the other below the middle of AB.

Step 3. Keeping the same radius, place the sharp point on B and draw two more arcs so that they cut the first arcs at P (above AB) and Q (below AB), as in fig-11. Each crossing is where an arc from A meets an arc from B.

Step 4. Join P to Q with a ruler (fig-12).

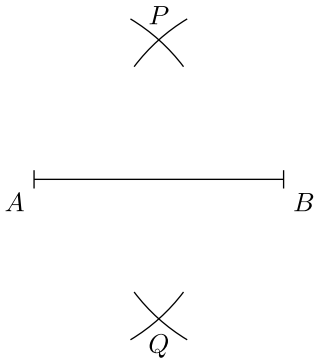
Step 5. The line PQ crosses AB at M. M is the midpoint of AB, so AM = MB, and PQ meets AB at 90°. Your segment is now bisected.

Visual



The same arcs with P joined to Q, the joining line crossing AB at its middle at M and carrying a right-angle mark in the upper right quadrant.

Visual



The segment AB with arc pairs crossing above and below it, the upper crossing labelled P and the lower one Q. Each crossing is drawn as a small X where an arc from A meets an arc from B.

Visual



The line segment AB, a stroke at each end, A on the left and B on the right.

Practice

1. When you bisect a line segment AB, the crossing line PQ meets AB at the point M. What does M being the midpoint of AB tell you about the two halves of the segment, and at what angle does PQ meet AB?
2. When constructing the bisection of segment AB, what must you do to the pair of compasses after drawing the two arcs from A, before you draw the arcs from B?
3. List, in order, the first three steps you take to bisect a line segment AB with a pair of compasses.

Section 4: Constructing a 90° Angle

A 90° angle is a right angle — the kind you see at the corner of a book or a sheet of paper. You can construct one exactly with your compasses and ruler, without measuring it. The trick is that a line drawn perpendicular to another line always makes a right angle where it meets it, and the compasses can help you find the point the perpendicular must pass through.

Here is the construction. You will need your compasses, your ruler and a pencil.

Step 1. Draw a straight line and label its ends A and B. Mark a point C on the line, at the place where you want the right angle to stand.

Step 2. Put the point of your compasses at C and draw an arc. This arc cuts the line at two points — label them D and E, one on each side of C.

Step 3. Now open your compasses **wider** than the distance DC from step 2. (The width you used in step 2 will not work here: arcs drawn with it from D and E would only meet again at C itself.) With this new, wider setting, place the compasses at D and draw an arc above the line.

Step 4. Keep the compasses at that same wider setting. Move them to E and draw another arc, cutting the first arc at a point F above the line.

Step 5. Use your ruler to join C to F with a straight line.

Step 6. The line CF is perpendicular to AB, so the angle where they meet is a right angle: $\angle FCB = 90\,^\circ$.

Let us follow the construction through once together, watching what each arc does.

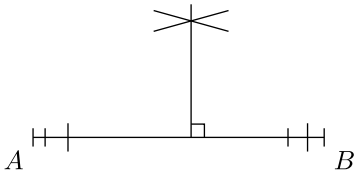
Draw line AB. Mark point C in the middle of it. With the compasses at C, draw an arc that cuts the line at two points. Call the left one D and the right one E — notice the arc has cut the line exactly twice, once at D and once at E.

Now widen the compasses. This is the key move: the new opening must be **larger** than the distance DC. With this wider opening, put the point at D and draw a long arc rising above the line. Then, without changing the compasses, put the point at E and draw another arc rising above the line. The two arcs cross each other above the line — that crossing point is F.

With your ruler, draw the straight line from C up to F. The line CF meets AB at a perfect right angle, and the little square you could draw at C between CF and CB is the right-angle mark. So $\angle FCB = 90\,^\circ$.

Check your own drawing: the two crossed strokes at the top of your figure mark where the two wider arcs met at F, and CF rises straight up from the line to meet it at a right angle.

Visual



A line with a stroke at each end, A on the left and B on the right, and four short strokes standing on the line, a pair towards A and a pair towards B. A perpendicular rises from a point near the middle, with a right-angle mark at its foot, and two short strokes cross each other near its top.

Practice

1. After you draw the arc from C that cuts the line at D and E, you must change the compasses before drawing the two arcs that meet at F. What change do you make, and why is it needed?
2. In this construction, the arc drawn with the compasses at C cuts the line AB at exactly two points. What are the two points labelled?
3. Having joined C to F with your ruler, state the size of $\angle FCB$ and the name given to the line CF in relation to AB.
4. Place these steps of the construction in the correct order by numbering them 1 to 5:
   (a) With the compasses at D at the wider setting, draw an arc above the line.
   (b) Draw a straight line AB and mark a point C on it.
   (c) With the compasses at E at the same wider setting, draw an arc cutting the first at F.
   (d) With the compasses at C, draw an arc cutting the line at D and E.
   (e) Join C to F with a ruler, so that $\angle FCB = 90\,^\circ$.

Section 5: Constructing a 60° Angle

A 60° angle can be drawn using only a ruler and a pair of compasses. The trick is that an equilateral triangle has three equal sides, and every angle in it is 60°. By swinging arcs of the same radius, you build two sides of an equilateral triangle, and the angle you open up between them is exactly 60°.

Here is the construction.

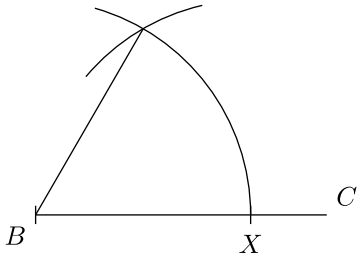
Step 1. Draw a straight line segment, say BC, with B at the left and C at the right. Put the point of your compasses on B and open them to any convenient radius. Draw an arc so that it cuts BC — mark where it cuts the line as X.

Step 2. Now put the point of your compasses on X, keeping exactly the same radius you used from B. Draw an arc so that it cuts the first arc. Call the point where the two arcs cross A.

Step 3. Draw the straight line from B to A.

The angle between the line BA and the line BC, measured at B, is 60°. Because you used the same radius for both arcs, BX and XA are equal in length, and BA equals them too — so the three points form an equilateral triangle, and an equilateral triangle has a 60° angle at every corner.

Visual



The line BC with B at the left, a stroke marking X part way along, a long arc swung from B through X and up to the left, a shorter arc swung from X cutting it, and a straight line from B to that crossing.

Practice

1. Follow the construction to draw a 60° angle at a point B on a straight line.
2. In the construction, why must the angle at B come out to exactly 60°?

Check your work

Attempt every question before you read this page.

Drawing a line parallel to a given line through a point

1. The constructed line through R is parallel to PQ: it runs side by side with PQ, the same distance apart all the way along, and never meets it.

   Place one edge of the set square exactly along PQ. Hold the ruler against the set square's other edge (the left-hand one if you are right-handed) so the set square cannot turn. Hold the ruler firm and slide the set square along it towards R, stopping when the edge that was on PQ passes through R. Draw a line along that edge through R. Since the set square slid without turning, the drawn line has the same direction as PQ and is therefore parallel to it.
2. The ruler sets the direction of the slide. Holding it firm means the set square can only move straight along it without turning, so the new edge keeps the same direction as the line it started on. If the ruler moved, the set square's slide would change direction and the drawn line would no longer be parallel.

Constructing Perpendicular Lines

1. $90\,^\circ$
2. The line XY you draw must cross AB at Q and meet it at a right angle; check with the right-angle corner of your set square that the two lines meet at $90\,^\circ$.
   a. Draw the line AB and mark the point Q on it. 2. Place the edge of the set square on AB so its base lies along the line, and rest the ruler against the hypotenuse of the set square. 3. Press firmly on the ruler, slide the set square along until its third (upright) edge touches Q, and draw the line XY along that edge. Since the set square's upright edge makes a right angle with its base, XY meets AB at $90\,^\circ$.

Bisecting a Line Segment

1. M is the midpoint, so AM = MB (the segment is divided into two equal halves), and PQ meets AB at 90°.
2. Keep the same radius (the compass opening) as you used from A, so that the arcs from B cut the first arcs at P and Q.
3. a. Open the compasses so the radius is about ¾ of the length of AB.
   b. Place the sharp point on A and draw two arcs, one above and one below the middle of AB.
   c. Keeping the same radius, place the sharp point on B and draw two arcs so they cut the first arcs at P and Q.

Constructing a 90° Angle

1. You open the compasses wider than the distance DC. The width used in step 2 would not work, because arcs drawn with it from D and E would meet only at C itself.
2. The two points are labelled D and E, one on each side of C.
3. $\angle FCB = 90\,^\circ$, and CF is perpendicular to AB.
4. (b) 1, (d) 2, (a) 3, (c) 4, (e) 5.

   First you draw the line and mark the point where the angle will stand — that is (b). Next, with the compasses at C, you draw the arc that cuts the line at D and E — that is (d). Then you widen the compasses and draw the first arc from D — (a). Keeping that wider setting, you draw the arc from E so it cuts the first at F — (c). Finally you join C to F and you have the right angle — (e).

Constructing a 60° Angle

1. The two arcs, swung from B and from X with the same radius, cross at A, and the angle ABC between the line BA and the line BC is 60°.
   a. Draw line segment BC. Put the compasses on B, open to any convenient radius, and swing an arc cutting BC at X.
   b. Keep the same radius, put the compasses on X, and swing an arc to cut the first arc. Mark the crossing A.
   c. Draw the straight line from B to A. The angle between BA and BC at B is 60°.
2. The two arcs cut at A, and drawing BA closes the shape ABX. All three sides — BX, XA and BA — are equal, so ABX is an equilateral triangle, and the angle at B is 60°.

   The compasses keep the same radius for both arcs, so the distance BX (from B to the cut on BC) and the distance XA (the second arc, swung from X) are equal. Since A lies on the first arc centred at B, BA equals the same radius as well. With BX = XA = BA, the triangle ABX has three equal sides — it is equilateral. Every angle in an equilateral triangle is 60°, so the angle at B is 60°.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-3ec527af	**Date**	
Subtopic	Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____